Mixed Residual Certificates for the Charged Penrose Construction

A source-specific Ram Calculus audit of manuscript M510

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Abstract

We examine the pinned charged spacetime Penrose manuscript at the level of its actual axial nondivergence equation, frozen divergence equation, coupled linearization, charged comparison, and original-data rigidity dependencies. Ram weak and residual formulations apply to these different operators in different spaces; they do not turn the coupled system into a symmetric coercive problem. We prove an explicit signed Morrey-source negative-norm budget, combine it with a two-component inverse criterion for the true linearized operator, and give a branch-preserving charged-area deficit certificate that has bounded sensitivity at the charge threshold. Exact algebraic and rational tests accompany the analysis. The additional results are local, conditional verification improvements: no geometric initial-data set, full nonlinear continuum solve, exhaustion limit, neutral Penrose theorem, or original-data rigidity theorem is numerically certified here. The distinction between the sharp upper-area bound and its restricted polynomial branch is maintained throughout. Contribution assessment: C2/5, source-specific mixed-source and branch certificates provide a modest verification extension, without a new global geometric theorem or a claim of historical novelty.

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1 Scope, pin, and mathematical conventions

The assigned public source is OpenAI, *The spacetime Penrose inequality with charge and original-data rigidity*, October 5, 2026, pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its relative directory in the study root is

`_sources/openai-math/preprints/The-spacetime-Penrose-inequality-with-charge-and-original-data-rigidity-October-5-2026/`.

The study root is

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`.

All eleven scientific section modules, including their proofs, were read in full. The assembly file, preamble, bibliography, and README were also read. Appendix A specifies the private passages actually read; routing maps are not theorem evidence. The existing upstream audit `_audit/upstream-20261008/` reports that the M510 selected files and edition links are unchanged at

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

That byte-identity audit is reused, not a fresh repository crawl and not an independent verification of any mathematical theorem. The assigned applicability row is L4. The contribution assessment C2 below is a separate axis and must not be inferred from L4.

1.1 Data, orientation, and the two inequality branches

The source uses $G = c = 1$, zero cosmological constant, signature $(-,+,+,+)$, and $K(U,V) = \mathbf{G}(\nabla_U n, V)$ for the future unit normal. Its exterior is a connected oriented smooth three-manifold with nonempty smooth compact boundary S, complete with its boundary included, and exactly one asymptotically Euclidean end. The inner normal ν points toward that end. Put $H = \operatorname{div}_S \nu$, $\theta_+ = H + \operatorname{tr}_S K$, and

$$2\mu = R_g + \tau^2 - |K|^2, \qquad J = \operatorname{div}(K - \tau g), \qquad \tau = \operatorname{tr} K, \tag{1.1}$$

$$\mu_m = \mu - (|E_f|^2 + |B_f|^2), \qquad J_m = J - 2(E_f \times B_f)^\flat. \tag{1.2}$$

Here E_f, B_f denote fields, to distinguish the ADM energy E. The hypotheses are $\mu_m \geq |J_m|$, $\operatorname{div} E_f = \operatorname{div} B_f = 0$, and $\mu, |J| \in L^1$. Smooth data need only satisfy $g - \delta = O_2(r^{-q})$, $K = O_1(r^{-1-q})$, $E_f, B_f = O_1(r^{-2})$, $q > 1/2$, with finite ADM and charge flux limits. These are not arbitrarily high derivative decay assumptions. Charge is normalized by $1/(4\pi)$ flux, energy by $1/(16\pi)$, and momentum by $1/(8\pi)$. Set $Q = (Q_E^2 + Q_B^2)^{1/2}$. Under metric changes the source retains the closed two-forms $\iota_{E_f}\mathrm{d}V_g, \iota_{B_f}\mathrm{d}V_g$, not fixed vector components.

A full cut is the entire intrinsic boundary of a connected closed smooth exterior containing the distant end. All its components and any contact with S count. The source defines

$$a_g(S) = \inf_{\Gamma \text{ full cut}} \operatorname{Area}_g(\Gamma), \qquad r_A = \sqrt{a_g(S)/(4\pi)}. \tag{1.3}$$

An arbitrary apparent-horizon component's area is not interchangeable with this infimum. Source `thm:numerical` assumes $\theta_+ \leq 0$ on all boundary components, or a fixed choice of $H \pm \operatorname{tr}_S K \leq 0$ on each component. It claims $E > |P|$, positive $m = \sqrt{E^2 - |P|^2}$, and

$$m \geq Q, \qquad r_A \leq m + \sqrt{m^2 - Q^2}. \tag{1.4}$$

It does not require connectedness or outermostness for this numerical inequality. Those hypotheses enter the separate rigidity theorem. For $0 < r_A \le Q$ the upper-area condition adds nothing to $m \ge Q$; only for $r_A > Q$ is it equivalent to

$$m \ge \tfrac{1}{2}(r_A + Q^2/r_A). \tag{1.5}$$

For example $(m, Q, r_A) = (1, 1, 1/2)$ satisfies (1.4), but the right side of (1.5) is $5/4$. This is an algebraic branch counterexample, not a constructed geometric exterior.

1.2 What this report proves

The proved statements below concern specified bounded domains, fixed coefficients or verified linearizations, signed source measures with quantified growth, and scalar charged-area algebra. Their proofs are independent of the source's global Penrose claim. Applying them to a particular filled construction still requires verified coefficient, boundary, inverse, and residual enclosures. No such geometric computation is supplied. Discussion of a source lemma means a dependency audit, not certification of all its proof steps.

2 Source-specific proof dependency audit

We retain the module labels so each claim can be checked in the pinned TeX. "Retained" below means that the source asserts and uses the result; it does not mean it has become a theorem of this report.

2.1 Full cuts and end preparation

Cuts. In `cuts.tex`, `ct:flux` carries the same total charge through every full cut by closedness and Stokes. The tubular calibration in `ct:full-cut-positive` establishes positivity of the infimum, without assuming a minimizing smooth cut exists. Confinement, finite-perimeter equivalence, strict compactness, and area differentiation are developed in `ct:confinement`, `ct:perimeter-equivalence`, `ct:strict-compactness`, and `ct:area-derivative`. They are needed again when the metric and obstacle vary. The compatible tangent-plane argument in `ct:compatible-planes` supports that variation. Replacing these by sampled surfaces misses topology, contacts, and nonattainment.

Weak end. In `end_preparation.tex`, the conformal and ADM flux identities precede `en:strictification`. Its scalar equation has a gradient-dependent drift; its linearized inverse cannot be justified by declaring the problem symmetric. The timelike reduction `en:timelike-reduction` uses a small conformal increase of the energy, fixed momentum and charge, and the already-proved timelike case. The null endpoint exclusion is thus a separate limiting argument, not a byproduct of a local elliptic residual estimate.

Supported inverses. The compact inverses `en:compact-inverses` require the appropriate divergence, affine scalar, and Killing-vector moments to vanish. The annular replacement `en:annular-replacement` exploits $q > 1/2$ and integrable sources. `en:rest-preparation` preserves flux forms, takes energy to m and momentum to zero, and compares the enclosing areas. It does not provide uniform compact-interior geometry for every member of the preparation sequence. A finite moment matrix cannot remove the continuous compatibility requirements or establish an inverse on its unresolved complement.

2.2 Filling, estimates, and the elliptic solver

Filling. In `filled_system.tex`, caps and a neck with length $O(1+N)$ close the inner boundary. The artificial data have $K = -\sigma L_0 g$ on the relevant filling region. Electromagnetic closedness is required on the original exterior, not automatically on the caps. `sy:differential`, `sy:scalar-equation`, `sy:positive-squares`, and `sy:coercivity` supply the exact identities and square estimates used later. Section 3 keeps their matrices rather than replacing them by a Laplacian.

A priori stage. In `a_priori.tex`, `es:height`, `es:floor`, and `es:distortion` give the height/floor/distortion bounds. The floor uses a stationary subtraction with coefficient $1/12$; the positive-square estimate must survive it. Constants polynomial in N at this stage are distinguished from arbitrary fixed-N higher regularity constants. In particular, a bound valid for every fixed N is not a bound uniform as $N \to \infty$.

Rough axial estimate. In `elliptic_solver.tex`, `so:rough-gradient` assumes bounded $z, \mathrm{d}z$, a bounded right side, $0 < c \le \chi \le 1$, controlled local geometry, and the special gradient-axis matrix. It obtains gradient Hölder control and a Hessian Morrey estimate $\int_{B_s} |\operatorname{Hess} z|^2 \le C s^{1+2\alpha}$. The proof uses a Calderon–Zygmund absorption near exponent two, a trace-reversed identity, a fixed-axis correction, and oscillation estimates. It does not differentiate a merely measurable χ. At constant Dirichlet faces the reflection and metric flattening are part of the argument. Our elementary axial residual bound below verifies one bounded-domain component, not this entire rough-gradient theorem.

Potential and coupling. `so:potential` uses a Green bound for a uniformly elliptic divergence operator and the growth $\mu(B_s) \le C_0 s^{1+\gamma}$, $0 < \gamma < 1$. It gives an L^∞ potential bound of order $r + r^\gamma$ for a bounded flux and such a measure on B_r. `so:coupled-regularity` combines the Hessian Morrey estimate with exponential transforms of Z, potential estimates, Harnack oscillation, and then Schauder interpolation. Its source includes $|\operatorname{Hess} f|^2$. It is not a globally differentiable map on $H^1 \times H^1$ in dimension three.

Existence. `so:trial-trace` solves the proper scalar nondivergence equation at trial Z. `so:compact-map` then freezes its coefficients and solves a symmetric divergence equation for a new Y. The full fixed-point derivative is not this frozen operator. `so:filled-solver` uses Leray–Schauder degree, not a contraction proof or a uniqueness statement for the coupled problem. Exhaustion $R \to \infty$ is at fixed N. The normalized end has exponential decay for f, $Z, t = O_j(r^{-1})$, improved integrable Δt, and integrable $u\Xi$. No validated numerical tail estimate is furnished by the degree argument alone.

2.3 Charged comparison and the neutral dependency

In `comparison.tex`, `cp:mass-cost` gives

$$E_{\widehat{g}} = E_* - \frac{1}{8\pi}\int u\Xi, \qquad \text{negative cost} \le P_N(\ell + \ell^{1/2}). \tag{2.1}$$

The height gap, a large conformal plateau, and perimeter compactness keep comparison cuts from exploiting the filling. The metric comparison with the original exterior must accompany any area estimate. A small local PDE error alone does not certify an infimum over full cuts.

After a constant electric–magnetic rotation, total charges are $(Q, 0)$. The projection square `cp:projection-square` contains

$$|X_e|^2 - |w \times X_e|^2 + |X_o + w \times X_e|^2. \tag{2.2}$$

The operation retains total charge, not prescribed charge on each individual boundary component. `cp:sharp-gradient` proves $\mathcal{T} \geq (3/4)|\mathrm{d}t|^2_{A_d}$ by a Schur minimization. Its sharper axial coefficient is

$$C(p,v) = \frac{(2p+1)(8+v)}{8+4v+6pv} \geq \frac{3}{4}, \qquad p \geq 0, \quad 0 \leq v < 1. \tag{2.3}$$

This formula belongs to the source; retaining it for coefficient budgets is not new mathematics of this report.

For $0 < s < 1$ the second scalar comparison uses profiles L, H_0, a lower height $y_0 = \log(1-s^2)$, and $Y = \mathcal{B}(y)$ with $\mathcal{B}'(y) = e^y L(y)$. In `cp:background-flux-equation` the actual equation is

$$2\,\mathrm{div}_{g_N}(A\nabla Y) = \mathrm{div}_{g_N}(h(Y)E_{\mathrm{ext}}), \qquad A = e^{t+\epsilon}\sqrt{D}\,A_d. \tag{2.4}$$

Here $h = H_0 \circ \mathcal{B}^{-1}$. Linearization in Y has flux $2A\nabla\zeta - h'(Y)E_{\mathrm{ext}}\zeta$. It is nonsymmetric, and E_{ext} is not presumed divergence free on the caps. The profile bound, asymptotic flux, and pure-trace comparison second fundamental form lead to the family

$$E_* \geq sQ + \tfrac{1}{2}(1-s^2)r_*, \qquad 0 < s < 1. \tag{2.5}$$

Crucially, `cp:numerical-prepared` invokes the separate OpenAI *The spacetime Penrose inequality and enclosing area*, September 27, 2026, Definitions 1.1–1.2 and Theorem 1.3. Its stated input is a smooth complete connected oriented one-end exterior with nonempty compact future-trapped boundary, neutral DEC, integrable constraint densities, finite future-timelike ADM four-momentum, positive full-cut area, and prepared O_6 metric/O_5 curvature decay with exponent in $(1/2, 1)$. Its output is the neutral bound $m \geq \frac{1}{2}\sqrt{a/(4\pi)}$. The presence of `cp:charged-comparison` does not license substituting a classical Riemannian charged theorem for this explicit dependency. The charged proof passes through exhaustion, N, ϵ, profile approximation, and end preparation limits with the source's order and dependencies retained. Endpoint $s \uparrow 1$ and optimization in s are additional passages. This report proves neither that neutral theorem nor the global charged theorem conditional on it in all analytic detail.

2.4 Why original-data rigidity remains separate

Source `thm:rigidity` adds: S connected, $\theta_+ = 0$, $\mathrm{Area}(S) = a_g(S)$, no interior full enclosing surface with $\theta_+ \leq 0$, and $m > Q$. Equality then concerns the entire original quadruple, including a smooth future-horizon or bifurcation-sphere embedding into dyonic Reissner–Nordström with the signed charges. Extremal equality is not covered by this statement.

The route in `variation.tex` is separation over active DEC rays and minimizing-cut measures (`va:separation`); Gaussian jets and boundary trace support; enlarged obstacles, including the mixed expansion case; distributional adjoint regularity (`va:smooth-moments`); and normalization to the ADM lapse–shift pair $b = (E/m, -P/m)$. `va:potentials` tests all compact closed two-form variations, not only exact forms. This is needed for global periods; simple connectedness is not an admissible shortcut. `va:stationary-data` yields the stationary metric with lapse u, shift X, $u \geq |X|$, and aligned null residual matter. `va:boundary` imposes $X = u\nu$ and the nonzero surface gravity $\kappa = (1 - Q^2/r_h^2)/(2r_h)$.

In `stationary_region.tex`, communication with the end, covariant jets without division by a vanishing lapse, smooth bifurcation attachment, stationary-end harmonic regularity, temporal graphs, and attachment causality precede `st:maximal-slice`. That step uses Chruściel–Wald and Bartnik, including the class-(b) bifurcate-boundary and causality hypotheses and the tilted-boundary limiting construction. A formal Killing initial-data equation by itself supplies none of those global conditions.

In `rigidity.tex`, `ri:staticity` obtains on the maximal slice an identity of the form

$$\int u'(|K'|^2 + 2|B'|^2 + h_*|X'|^2) = 0. \tag{2.6}$$

The sign, boundary flux, potential constants, and normalization are essential. `ri:static-mass` identifies the original invariant ADM mass. `ri:rn-identification` applies electrostatic uniqueness only after the full static electro-vacuum system and its boundary/asymptotic normalization hold. `ri:original-embedding` must then recover the original slice, the regular advanced-time horizon limit, orientation, normal, and both electromagnetic fields. `ri:sharp-examples` checks admissible static and boosted examples. Residual control of a comparison metric does not certify any of these original-data identifications.

2.5 External dependency inventory

The complete bibliography and all scientific modules were inspected for dependency routing. The following inventory distinguishes proof inputs from context. External originals are not all re-proved or fully audited.

1. BV compactness, lower semicontinuity, coarea and approximation: Ambrosio–Fusco–Pallara, Theorem 3.23, Proposition 3.6, Theorem 2.39, Theorem 3.46 and Remark 3.50; strict measure continuity also relates to Spector's Reshetnyak paper, listed in the source bibliography.

2. Obstacle and minimizing-boundary regularity: Huisken–Ilmanen, Regularity Theorem 1.3(iii). Outermost MOTS input in comparison: Andersson–Metzger, Theorem 5.1. These are not ordinary PDE residual lemmas.

3. Supported de Rham inverses: Costabel–McIntosh, Theorems 3.2 and 4.9 and Corollary 4.11, with the source's additional moment handling.

4. Geometric Sobolev step: Nash, Theorem 2, p. 59; Michael–Simon; Simon's GMT notes, Chapter 4, Section 6, Theorem 6.7.

5. Elliptic estimates, maximum principles and degree: Gilbarg–Trudinger, Chapters 6, 8, 9, 11, specifically Theorems 9.9 and 8.18; the source itself supplies the special axial and coupling arguments.

6. Rough divergence Green bound: Littman–Stampacchia–Weinberger, Theorem 7.1 and equation (7.9). The signed negative-norm theorem below instead proves its own Euclidean Newton-potential bound.

7. The separate neutral enclosing-area theorem, Definitions 1.1–1.2 and Theorem 1.3, is a substantive global dependency of the numerical charged conclusion, not a previously certified step here.

8. Maximal slices and their causality setup: Chruściel–Wald, Definitions 2.1, 2.3, 2.4, equation (4.1), Theorems 3.2 and 4.2, Proposition 4.2, Lemma 4.2 and Corollary 3.6; Bartnik, Theorems 3.1, 4.2 and 5.3 with the barrier and tilt estimates specified in `stationary_region.tex`.

9. Staticity comparison: Sudarsky–Wald, equation (40) and Theorem 2. Electrostatic identification: Borghini–Cederbaum–Cogo, Theorem 3.1 and Proposition 6.1. The source must verify their hypotheses before use.

10. Historical/formulation context: Penrose 1973; Bray's Riemannian proof; Bray–Khuri's Jang approach; Ben-Dov and Carrasco–Mars on pitfalls in apparent-horizon formulations; Khuri–Weinstein–Yamada on the charged Riemannian case. Context is not a substitute for the spacetime theorem.

3 The actual elliptic operators

Fix the filling parameter N and a bounded smooth outer domain. All constants below may depend on that domain, N, and the candidate's coefficient bounds. Source notation is retained except that v below is the scalar speed and φ denotes an increment. Put

$$p(t) = N\vartheta(Nt), \quad l(t) = \exp\left(\int_0^t p(s)\mathrm{d}s\right), \quad \ell = e^{-N\epsilon}, \quad \eta = \ell^{3/2}, \tag{3.1}$$

$$D = 1 + l^2|\mathrm{d}f|^2, \quad Z = t + \tfrac{1}{4}\log D, \quad w = lD^{-1/2}\nabla f, \quad v = |w|^2, \quad d = D^{-1} = 1 - v, \tag{3.2}$$

$$q = 2 + pv, \quad \chi = 2d/q, \quad A_d = I - w \otimes w, \quad A_\chi = I - \frac{2+p}{q} w \otimes w, \tag{3.3}$$

$$u = e^t l\sqrt{D}, \quad H_f = lD^{-1/2}\operatorname{Hess} f, \quad C = K + H_f, \quad F = \operatorname{tr}_{A_\chi} C. \tag{3.4}$$

For $w \neq 0$, $A_\chi = I - (1-\chi)e \otimes e$, $e = w/|w|$. Its eigenvalues are $(1, 1, \chi)$, while those of A_d are $(1, 1, d)$. At $w = 0$ the first formula in (3.3) avoids an undefined axis. The floor/distortion bounds provide $d > 0$ and hence $\chi > 0$ at fixed parameters; a numerical claim still needs a quantified lower bound.

The differential identity and scalar-curvature identity are

$$2\mathrm{d}Z = q\mathrm{d}t + H_f(w, \cdot), \tag{3.5}$$

$$\tfrac{1}{2}e^{2t}R_{\widehat{g}} = \mu + J(w) + \mathcal{T} + w(F) - F\tau - u^{-1}\operatorname{div} V, \tag{3.6}$$

$$V = uA_\chi(2\nabla Z + K(w, \cdot)), \qquad \widehat{g} = e^{2t}(g + l^2\mathrm{d}f^2). \tag{3.7}$$

The source's square decomposition yields

$$|T_{\mathrm{block}}|^2 + |M_{\mathrm{block}}|^2 + dj^2 + F^2 + |\mathrm{d}t|^2_{A_d} \leq 20(1+N)^2\mathcal{T}. \tag{3.8}$$

It is a tensor-algebra estimate, not coercivity of the coupled Jacobian.

3.1 Nondivergence trace equation

For the homotopy, replace K by $K_a = aK$ and recompute the quantities. Write $\omega = \sqrt{D}/l$. The normalized first equation is

$$\mathcal{F}_1(f, Z) = A_\chi : \operatorname{Hess} f + \omega \operatorname{tr}_{A_\chi} K_a - \omega\eta f = 0. \tag{3.9}$$

At fixed trial Z, its derivative has the form $A_\chi : \operatorname{Hess}\varphi + b \cdot \nabla\varphi - c_0\varphi$ with $c_0 = \omega\eta > 0$, and bounded drift at a smooth candidate. It is a proper nondivergence operator. In general

$$A : D^2\varphi = \operatorname{div}(A\nabla\varphi) - (\operatorname{div} A) \cdot \nabla\varphi \tag{3.10}$$

only when the coefficient derivatives exist in a sufficient sense. For measurable A the right side need not define the same H^{-1} problem. Dropping the coefficient-derivative term is an operator change.

3.2 Frozen weak equation and the full derivative

The second source equation is $\operatorname{div} V = bu\Xi$. Define

$$M = 2uA_\chi, \quad J_0 = uA_\chi K_a(w, \cdot), \qquad \langle \mathcal{F}_2, \psi\rangle = \int (M\nabla Z + J_0) \cdot \nabla\psi + \int bu\Xi\psi. \tag{3.11}$$

Here Ξ includes $\delta_0(\mathcal{T} + \eta a_w|\mathrm{d}f|)$, a positive source ρ, and the height-cutoff penalty $-\vartheta(N(t + \epsilon))C_N(\rho_0 + v\rho_1)$. Thus its differentiation must retain the cutoff, tensor-square, and gradient

terms. The speed $a_w = |w|$ is not the homotopy parameter a; the product $a_w|\mathrm{d}f| = lD^{-1/2}|\mathrm{d}f|^2$ is smooth even at zero slope. In the compact map only ∇Y is unfrozen, so solving for Y gives the genuinely symmetric principal block $-\operatorname{div}(M\nabla Y)$.

For variations $(\varphi, \zeta) = (\delta f, \delta Z)$ at fixed geometry, direct differentiation of (3.2) gives

$$\dot{t} = \frac{2\zeta - l^2 D^{-1}\langle \mathrm{d}f, \mathrm{d}\varphi\rangle}{q}. \tag{3.12}$$

Dots below mean total variations, including this dependence, $\dot{p} = p'\dot{t}$, $\dot{l} = pl\dot{t}$, and $\dot{D}$. In particular, source `sy:penalty` and the sentence after `sy:normalized-trace` (`filled_system.tex`, lines 542–569) give the smooth product $S_f = \eta a_w|\mathrm{d}f| = \eta l D^{-1/2}|\mathrm{d}f|^2$, whose derivative is

$$\dot{S}_f = \eta l D^{-1/2}\left[2\langle \mathrm{d}f, \mathrm{d}\varphi\rangle + \left(p\dot{t} - \frac{\dot{D}}{2D}\right)|\mathrm{d}f|^2\right]. \tag{3.13}$$

It equals zero at $\mathrm{d}f = 0$ and extends smoothly there. Differentiating a_w and $|\mathrm{d}f|$ separately at zero is neither needed nor justified. No norm-source regularization or nonvanishing-gradient assumption is being substituted for the source equation. This check does not waive the strong-space admissible-neighborhood requirement for a nonlinear Fréchet certificate discussed below. No coefficient is silently frozen:

$$\begin{aligned} D\mathcal{F}_1[\varphi, \zeta] &= A_\chi : \operatorname{Hess}\varphi - \omega\eta\varphi \\ &\quad + \dot{A}_\chi : (\operatorname{Hess} f + \omega K_a) + \dot{\omega}(\operatorname{tr}_{A_\chi} K_a - \eta f), \end{aligned} \tag{3.14}$$

$$\begin{aligned} \langle D\mathcal{F}_2[\varphi, \zeta], \psi\rangle &= \int (M\nabla\zeta + \dot{M}\nabla Z + \dot{J}_0)\cdot\nabla\psi \\ &\quad + \int b(\dot{u}\Xi + u\dot{\Xi})\psi. \end{aligned} \tag{3.15}$$

The $\dot{\Xi}$ terms include $\operatorname{Hess}\varphi$ and $\nabla\zeta$. Equations (3.14)–(3.15) expose why the two-field linearization is not the symmetric frozen solve.

At a smooth candidate with bounded coefficient derivatives on a smooth bounded domain, the displayed linear map is bounded on

$$X = (H^2 \cap H_0^1) \times H_0^1, \qquad Y = L^2 \times H^{-1}. \tag{3.16}$$

Indeed the first component contains bounded coefficients times second derivatives of φ, first derivatives of φ, and ζ. The second is a divergence of L^2 vector fields plus L^2 terms, using $\|\psi\|_2 \le C_P\|\nabla\psi\|_2$. This bounded linearization is not a claim that the original nonlinear map is continuously Fréchet differentiable on all of X. In three dimensions H^2 does not bound ∇f in L^∞; the quadratic Hessian source creates additional product-space issues. Nonlinear validation must use a stronger admissible neighborhood, such as suitable Hölder spaces, and verify its actual mapping and inverse.

4 A bounded-domain axial residual certificate

This elementary result isolates a legitimate nondivergence use of a Ram residual: the norm is a strong L^2 residual controlling a Hessian, not the weak energy norm of an invented divergence operator.

Theorem 4.1 (Axial inverse at fixed ellipticity)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded convex $C^{1,1}$ domain for which the Dirichlet Laplacian maps $H^2 \cap H_0^1$ onto L^2 and satisfies $\|D^2 z\|_2 \le \|\Delta z\|_2$. Let*

$e(x)$ be a measurable unit field and $c \le \chi(x) \le 1$, $c > 0$. Set $A = I - (1-\chi)e \otimes e$. Then $Lz = A : D^2 z$ is an isomorphism from $H^2 \cap H_0^1$ to L^2 and

$$\|D^2 z\|_2 \le c^{-1} \|Lz\|_2. \tag{4.1}$$

For an exact solution $Lz = f$ and an approximation z_h with the same Dirichlet trace, replace z in the left side by $z - z_h$ and Lz on the right by $f - Lz_h$.

Proof. The Hilbert norm $\|D^2 z\|_2$ is complete and equivalent to the H^2 norm on this Dirichlet space. The map $Tz = \Delta^{-1}(f + (I - A) : D^2 z)$ obeys

$$\|D^2(Tz - T\widetilde{z})\|_2 \le \|(I - A) : D^2(z - \widetilde{z})\|_2 \le (1 - c)\|D^2(z - \widetilde{z})\|_2.$$

Banach's theorem gives a solution for every f. For any z, $\|D^2 z\|_2 \le \|\Delta z\|_2 \le \|Lz\|_2 + (1 - c)\|D^2 z\|_2$. Absorption proves the estimate and injectivity. Apply it to the error. □

The rank-one identity $\|I - A\|_F = 1 - \chi$ is the entire coefficient input. No derivative of χ or of the axis was used. If a chart or coefficient enclosure gives $\|I - \widetilde{A}\|_{F,\infty} \le \theta < 1$, the same proof gives $(1 - \theta)^{-1}$. Metric first-order terms, drift, and the zero-order term of (3.9) still have to be included in a perturbation/inverse budget; they have not vanished. Nonconvex domains require a different Laplace regularity constant and geometry argument. An inhomogeneous trace error requires a verified H^2 lift b and the bound applied to $z - z_h - b$, with Lb in the residual.

Remark 4.2 (No novelty claim for the axial theorem). This is a special near-Laplacian/Cordes contraction argument, consistent with established nondivergence numerical analysis [7] and the source's fixed-axis correction. It does not prove its entire Morrey regularity step, and it is not counted alone as a new contribution. The constant deteriorates as $c \downarrow 0$, as the construction permits when its parameters degenerate.

Even a smooth proper scalar equation need not have a coercive unweighted energy form. On $(0, 1)$ let $Lv = -v'' + 24xv' + v$ with zero endpoints. For $v = x(1 - x)$, integration gives

$$\int_0^1 vLv = \int_0^1 |v'|^2 - 11 \int_0^1 v^2 = \tfrac{1}{3} - \tfrac{11}{30} = -\tfrac{1}{30}. \tag{4.2}$$

The positive zero-order coefficient permits a maximum-principle uniqueness argument, but not the asserted positive energy inequality. This is an illustrative operator-class counterexample, not a claim that these particular coefficients occur in the source geometry.

5 Signed Morrey sources and frozen weak error

The source's potential estimate concerns a nonnegative control measure. A computed residual may be signed and may have both a divergence part and a concentration part. The following extension gives a quantitative negative-norm budget using total variation; signed mass cancellation alone is insufficient.

Theorem 5.1 (Explicit signed-source budget)**.** *Let $\Omega \subset \mathbb{R}^3$ be bounded and let ν be a finite signed measure supported in $\overline{\Omega}$, interpreted on zero extensions of $H_0^1(\Omega)$. Put $\mu = |\nu|$ and $M_0 = \mu(\mathbb{R}^3)$. Suppose for all centers in $\mathbb{R}^3$ and $0 < s \le h_0$,*

$$\mu(B_s(x)) \le K_0 s^{1+\gamma}, \qquad 0 < \gamma < 1. \tag{5.1}$$

Then, for $0 < h \le h_0$,

$$\|\nu\|_{H^{-1}(\Omega)} \le \mathcal{N}(h) := \left[\frac{M_0}{4\pi}\left(\frac{M_0}{h} + \frac{K_0 h^\gamma}{\gamma}\right)\right]^{1/2}, \tag{5.2}$$

where H_0^1 has norm $\|\nabla \cdot\|_2$. For $M_0, K_0 > 0$ the best h in this bound is $\min\{h_0, (M_0/K_0)^{1/(1+\gamma)}\}$.

Proof. With $G(x) = 1/(4\pi|x|)$, layer integration gives, uniformly in x,

$$(G * \mu)(x) \le \frac{1}{4\pi}\left[\frac{M_0}{h} + \int_0^h \frac{\mu(B_s(x))}{s^2}\mathrm{d}s\right] \le \frac{1}{4\pi}\left(\frac{M_0}{h} + \frac{K_0 h^\gamma}{\gamma}\right).$$

The Newton energy is consequently at most M_0 times that last bound. For smooth compactly supported approximations the identity $\int |\nabla(G*\nu)|^2 = \iint G(x-y)\mathrm{d}\nu(x)\mathrm{d}\nu(y)$ and Cauchy–Schwarz imply $|\int \phi \mathrm{d}\nu| \le \mathcal{N}(h)\|\nabla\phi\|_2$. To pass to measures, mollify with a nonnegative approximate identity. The corresponding positive measures still have total mass M_0 and the same all-center growth bound, because their ball masses are averages of translated ball masses. The uniform estimate passes to smooth test functions by weak convergence, and then extends by density to H_0^1. This also defines the action unambiguously; arbitrary pointwise representatives of Sobolev functions are not being used. Finally the derivative of $M_0/h + K_0 h^\gamma/\gamma$ is $-M_0/h^2 + K_0 h^{\gamma-1}$. It changes sign at the stated unconstrained minimum, proving the constrained choice. The zero measure is trivial. □

The all-center condition is deliberate. A growth estimate only at sampled centers, or only for the signed measure ν instead of $|\nu|$, does not satisfy the theorem. On a Riemannian chart the pushforward measure, metric norm equivalence, and volume factors must be enclosed; one cannot reuse the Euclidean $1/(4\pi)$ without those conversions. For example if $\int |\nabla_g \phi|^2 \mathrm{d}V_g \ge \lambda_g \int |D\phi|^2 \mathrm{d}x$, the Euclidean bound is multiplied by $\lambda_g^{-1/2}$ for the geometric dual norm.

Corollary 5.2 (Mixed residual and frozen coefficient defect)**.** *Let $M(x)$ be measurable symmetric and $\lambda I \le M \le \Lambda I$ on a bounded Lipschitz domain. Suppose $-\operatorname{div}(M\nabla z) = S$ with specified Dirichlet data. Choose an approximation z_h with the correct trace after a lift, and suppose the full weak residual has a decomposition*

$$r = S + \operatorname{div}(M\nabla z_h) = f + \operatorname{div} B + \nu, \quad f \in L^2, \quad B \in L^2(\Omega;\mathbb{R}^3),$$

with ν as in Theorem 5.1. Then

$$\|r\|_{H^{-1}} \le C_P\|f\|_2 + \|B\|_2 + \mathcal{N}(h) =: \mathcal{R}, \tag{5.3}$$

$$\|\nabla(z - z_h)\|_2 \le \lambda^{-1}\mathcal{R}, \qquad \|z - z_h\|_M \le \lambda^{-1/2}\mathcal{R}. \tag{5.4}$$

If a computed coefficient M_h was used, add $\|(M - M_h)\nabla z_h\|_2$ to the vector-field budget. If the trace correction is b, also include $\|M\nabla b\|_2$ when evaluating the residual of $z_h + b$, and retain $\|b\|_M$ for an error relative to the uncorrected candidate.

Proof. For $\|\nabla\psi\|_2 = 1$, the three terms act as $\int f\psi$, $-\int B\cdot\nabla\psi$, and $\langle\nu,\psi\rangle$. Poincare, Cauchy–Schwarz and Theorem 5.1 give (5.3). The weak error identity is $\int M\nabla e\cdot\nabla\psi = \langle r,\psi\rangle$. Test by e, use ellipticity, and divide. The coefficient and lift terms follow by adding their actual flux defects, with the triangle inequality. □

The symmetric hypothesis here is valid for the *frozen* matrix $M = 2uA_\chi$ in (3.11). It is not asserted for (3.14)–(3.15). For a divergence residual, an H^{-1} bound avoids differentiating a rough flux into a possibly unavailable L^2 source. The decomposition is not unique: one can minimize the certified right side over admissible decompositions, but must account for all components and quadrature errors.

The related Ram hypercircle passage applies when its symmetric elliptic realization, flux sign, and equilibration hypotheses hold. For $q = -M\nabla z$, the convention is $\operatorname{div} q = S$. An admissible $H(\mathrm{div})$ flux has normal continuity; independent cellwise fluxes do not automatically qualify. A singular measure source need not be in L^2, so one should use the dual residual statement above instead of silently applying an L^2-source hypercircle identity.

6 A mixed inverse test for the true linearization

The source's coupled degree argument is not an inverse bound. The next conditional theorem gives a usable alternative gate at one smooth candidate without changing its Jacobian.

Theorem 6.1 (Two-component inverse budget)**.** *Let X_i, Y_i be Banach spaces and $D_i : X_i \to Y_i$ bounded isomorphisms. Write the actual linear operator as $L = D + E$, $D = \operatorname{diag}(D_1, D_2)$, and assume*

$$\|D_i^{-1} E_{ij} x_j\|_{X_i} \le k_{ij} \|x_j\|_{X_j}, \qquad K = (k_{ij}) \ge 0.$$

If

$$k_{11} < 1, \quad k_{22} < 1, \quad \Delta_K = (1 - k_{11})(1 - k_{22}) - k_{12} k_{21} > 0, \tag{6.1}$$

then $L : X_1 \times X_2 \to Y_1 \times Y_2$ is an isomorphism. For $Lx = r$, let $b_i \ge \|D_i^{-1} r_i\|_{X_i}$. Componentwise,

$$\begin{pmatrix} \|x_1\| \\ \|x_2\| \end{pmatrix} \le \frac{1}{\Delta_K} \begin{pmatrix} 1 - k_{22} & k_{12} \\ k_{21} & 1 - k_{11} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}. \tag{6.2}$$

Proof. The conditions make $(I - K)^{-1}$ entrywise nonnegative and the Perron root of K strictly less than one. More explicitly set $t = (I - K)^{-1}(1, 1)^T$, so $t_i \ge 1$ and $(Kt)_i = t_i - 1$. In the weighted maximum norm $\max_i \|x_i\|/t_i$, $D^{-1}E$ has norm at most $\max_i(1 - 1/t_i) < 1$. The equation $x = D^{-1}r - D^{-1}Ex$ therefore has a unique solution for every r, giving surjectivity as well as injectivity. Iterating its componentwise bound gives $\sum_{n \ge 0} K^n b = (I - K)^{-1} b$, which is (6.2). □

For M510, take the mixed spaces (3.16). One legitimate choice is a separately verified full scalar trace inverse for D_1 and the frozen divergence inverse for D_2. Every remaining derivative term in (3.14)–(3.15), including self-coupling, belongs in E. Alternatively use the axial inverse in Section 4 as D_1 and budget its drift and reaction in E_{11}. Neither choice guarantees (6.1). Strong coupling, large cutoffs, small χ, or the domain's Poincare constant may cause failure. Failure means this sufficient certificate is inconclusive, not that the source solution fails to exist.

The source-specific improvement is the combination: strong axial L^2 residual in component one; signed mixed-source H^{-1} budget in component two; actual cross derivatives; and explicit inverse amplification. For example, the exact rational enclosure

$$K = \begin{pmatrix} 1/10 & 1/5 \\ 1/4 & 1/10 \end{pmatrix}, \quad b = \begin{pmatrix} 1/1000 \\ 1/500 \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} \|x_1\| \\ \|x_2\| \end{pmatrix} \le \begin{pmatrix} 13/7600 \\ 41/15200 \end{pmatrix} \tag{6.3}$$

has $\Delta_K = 19/25$. These numbers are a manufactured certificate test, not measured constants for a filled spacetime exterior.

6.1 Nonlinear and finite-coordinate limits

For a genuinely C^1 map $\mathcal{F} : U \subset \mathcal{X} \to \mathcal{Y}$ on stronger verified spaces, a bounded injective preconditioner $B : \mathcal{Y} \to \mathcal{X}$ and a complete ball may yield the standard test

$$\|B\mathcal{F}(x_0)\| \le \eta_0, \qquad \sup_{x \in \overline{B}_r(x_0)} \|I - B\mathcal{F}'(x)\| \le q_0 < 1, \quad \eta_0 + q_0 r \le r. \tag{6.4}$$

Indeed $T(x) = x - B\mathcal{F}(x)$ maps the ball into itself and is a contraction; injectivity of B converts its fixed point to a zero. This yields local existence and uniqueness in that ball, with distance at most $\eta_0/(1 - q_0)$. It is a classical certificate and not a novelty claim. The report has not verified (6.4) for the source.

An invertible finite Ram coordinate transform preserves a finite trial space, not the infinite-dimensional inverse. For a test basis ψ_i in H_0^1, use the physical Gram matrix $G_{ij} = \int \nabla\psi_j \cdot \nabla\psi_i$ and moments $b_i = \langle r, \psi_i \rangle$. The restricted dual norm is $\sqrt{b^* G^{-1} b} \le \|r\|_{H^{-1}}$; it is a lower diagnostic, not an upper certificate. An unresolved orthogonal residual may be nonzero even if every stored moment vanishes. A finite-rank inverse is not injective on an unrestricted continuum residual space. Neither an algebraic condition number nor successful moment fitting resolves that defect. Boundary lifts, flux continuity, coefficient error, quadrature and the complement remain explicit obligations.

7 Branch-preserving charged-area certificates

We now treat the last scalar step of the source, independently of the geometric proof that produces its support family.

Theorem 7.1 (Support envelope and stable deficit)**.** *For $r > 0$, $Q \ge 0$, set*

$$\Phi(r, Q) = \sup_{0 \le s \le 1} \left[sQ + \tfrac{1}{2}(1 - s^2)r \right] = \begin{cases} Q, & r \le Q, \\ (r + Q^2/r)/2, & r > Q. \end{cases} \tag{7.1}$$

For $m \ge 0$, the assertion $m \ge \Phi(r, Q)$ is equivalent to $m \ge Q$ together with $r \le m + \sqrt{m^2 - Q^2}$. Moreover for any two such pairs,

$$|\Phi(r, Q) - \Phi(\widetilde{r}, \widetilde{Q})| \le \tfrac{1}{2}|r - \widetilde{r}| + |Q - \widetilde{Q}|. \tag{7.2}$$

Consequently a candidate deficit $\widetilde{m} - \Phi(\widetilde{r}, \widetilde{Q})$ with absolute input errors $\delta_m, \delta_r, \delta_Q$ differs from the true deficit by at most $\delta_m + \delta_r/2 + \delta_Q$.

Proof. The profile is a concave quadratic in s, with derivative $Q - rs$. The maximizer is $s_* = \min(1, Q/r)$, which gives (7.1). If $r \le Q$ and $m \ge Q$, then $r \le m \le m + \sqrt{m^2 - Q^2}$. For $r > Q$, the quadratic inequality $r^2 - 2mr + Q^2 \le 0$ is equivalent to $m \ge (r + Q^2/r)/2$, and to lying between the two roots when $m \ge Q$. Its lower root is at most $Q < r$, so only the upper root adds a restriction. Conversely the upper root and $r > Q$ put r between the roots. For each $s \in [0, 1]$ the coefficient of r is between zero and $1/2$, and that of Q between zero and one. The supremum of these profiles thus satisfies (7.2) by comparison in both directions. Adding the mass error proves the last assertion. □

Unlike differentiating $m + \sqrt{m^2 - Q^2}$, this deficit estimate has no coefficient singularity at extremality. It does not remove the sensitivity of recovering m from nearly null ADM energy–momentum. For independent enclosures $E \in [E_-, E_+]$, $|P| \in [P_-, P_+]$, first require $E_- > P_+ \ge 0$ and then use

$$\sqrt{E_-^2 - P_+^2} \le m \le \sqrt{E_+^2 - P_-^2}. \tag{7.3}$$

Evaluate square roots with directed rounding or rational square brackets. If the input interval crosses the null threshold, certification of positive invariant mass is unresolved. The charge norm is one-Lipschitz in its two signed flux components, so their Euclidean error bound controls δ_Q. The area-to-radius conversion needs a positive lower area enclosure; it is not uniformly Lipschitz at zero.

For intervals, monotonicity of Φ gives the particularly direct gate

$$m_- \ge \Phi(r_+, Q_+) \quad \Longrightarrow \quad \text{the enclosed triple satisfies the sharp scalar bound.} \tag{7.4}$$

Only if $r_- > Q_+$ may the entire interval be treated as the polynomial branch. If intervals cross $r = Q$, use (7.1), not a branch selected by the midpoint. These statements are algebraic enclosures; they do not certify that the measured r is the full enclosing radius.

Proposition 7.2 (Finite profiles with a rigorous remainder)**.** *Let* $0 = s_0 < \cdots < s_n = 1$*, with maximum spacing* h*. If* $r > Q$ *and* $s_* = Q/r$*, then*

$$0 \le \Phi(r, Q) - \max_i \{s_i Q + (1 - s_i^2) r/2\} \le r h^2/8.$$

For $r \le Q$ *the loss is zero because* $s_n = 1$*. If only* $s_n = 1 - \kappa$ *is available and* $r \le Q$*, the endpoint loss is exactly* $\kappa(Q - r) + r\kappa^2/2$*. If valid lower bounds are* $m \ge s_i Q + (1 - s_i^2) r/2 - \delta_i$*, then they imply* $m \ge \Phi(r, Q) - \max_i \delta_i - \varepsilon_{\mathrm{grid}}$ *with the applicable grid loss above.*

Proof. For an interior maximum the exact difference is $\Phi - F_s = r(s - s_*)^2/2$. A nearest node is at distance at most $h/2$. The endpoint assertions follow by substituting $s = 1 - \kappa$. Finally $\max_i (F_{s_i} - \delta_i) \ge \max_i F_{s_i} - \max_i \delta_i$. □

The source proves its profile inequalities for $0 < s < 1$ and obtains endpoint information by a limit. A computational grid containing 1 is not a substitute for that limit; an endpoint must be independently justified or the displayed endpoint loss retained. An approximate bound $m \ge \Phi - \delta$ is not the exact Penrose inequality when $\delta > 0$. Neither zero computed defect nor a small positive interval proves original-data rigidity; that would require the exact equality and all the geometric hypotheses audited above.

8 Validation design and evidence

The accompanying `validate_M510.py` performs exact symbolic or rational checks using SymPy, writing machine-readable and readable receipts in the owned audit area. It does not discretize a black-hole exterior. The tests are organized to falsify incorrect reformulations, not merely to confirm a few favorable floating-point examples.

1. Verify eigenvalues of the axial matrix and the implicit $\dot{t}$ identity, retaining both height and gradient increments.
2. Verify the nonnegative numerator of $C(p, v) - 3/4$ and the monotonicity of C in p. This is a check of the quoted scalar Schur coefficient, not a replay of the full tensor square decomposition.
3. Evaluate (4.2) exactly. This rejects the false implication "proper uniformly elliptic implies symmetric coercive."
4. Check the rational block inverse and its component bounds in (6.3); include a singular threshold case with $k_{12}k_{21} = 1$, $k_{11} = k_{22} = 0$.
5. For a radial measure density $|x|^{\gamma-2}$ on the unit ball, check the exact solution of $-\Delta z = |x|^{\gamma-2}$, $z = (1 - |x|^\gamma)/(\gamma(1 + \gamma))$, and its energy $4\pi/[(1 + \gamma)^2(1 + 2\gamma)]$. At $\gamma = 1/2$ it is bounded and in H_0^1 despite the singular source. This illustrates the negative-source regime. It is not a proof of all-center growth for an arbitrary computed measure; that remains a hypothesis.
6. Check both branches, the explicit false polynomial example, exact rational profile-grid losses, and rational input perturbations against (7.2). A finite sweep is a regression test; the theorem's universal guarantee comes from its proof, not sampling.
7. Check that finite tested moments can vanish for a nonzero residual, and that a physical Gram metric, not raw coefficient norm, is needed for the restricted dual norm.

The receipts distinguish exact identities, exact finite enumerations, and unperformed geometric validation. No Monte Carlo confidence claim, benchmark speedup, empirical continuum convergence rate, certified ADM tail, or validated minimal surface is reported.

The provenance script hashes the exact report, scripts, consulted inputs and final PDF; the assessment uses the report's final TeX SHA. Build receipts record native editor/compiler availability separately from the physical MiKTeX export. Rendered-page inspection is a formatting check, not a proof check.

9 Assessment, advantages, and limitations

Assessment C2/5. The report supplies a source-specific verification extension by combining (3.14)–(3.15), the signed-source budget, mixed inverse amplification, and the stable upper-area envelope. The signed-source proof makes the total-variation, scale and coordinate requirements explicit; the branch test avoids a genuinely false global polynomial formulation. The standard residual, Cordes and contraction theorems themselves are not novel. No literature priority is claimed for their combination. Without a verified geometric candidate and with the source's global dependencies retained, C3 would overstate the demonstrated advance. L4 applicability does not raise C2.

Advantages. The formulation respects the correct principal symbols, includes mixed and negative sources, handles trace and coefficient mismatch, exposes coupling constants, and gives a transparent failure mode when a sufficient inverse test is too weak. The scalar charged certificate is stable at the branch junction and records finite profile and flux uncertainties. Exact tests are reproducible without large numerical infrastructure. The report also prevents comparison metric rigidity from being mistaken for original-data rigidity.

Costs and disadvantages. The total-variation measure bound can be pessimistic when cancellation is substantial. Obtaining an all-center Morrey enclosure is harder than fitting moments. The block test may fail despite invertibility, especially when χ is small or the neck is long. A physical Gram matrix can be ill-conditioned and still omits the complement. Strong-space nonlinear validation is more expensive than a weak residual calculation. Exterior truncation, boundary flux, cut infima, and parameter limits all add error budgets that have not been supplied here. None of these deficiencies is cured by a change of polynomial coordinates.

Public literature boundary. A targeted public-title search on October 8, 2026 checked primary records for the charged Riemannian theorem [3], rough divergence Green theory [4], maximal slices [5], electrostatic uniqueness [6], and Cordes nondivergence discretization [7]. The latter explicitly provides a priori and a posteriori numerical analysis, so a generic nondivergence residual estimate cannot be advertised as new. The Riemannian charged result concerns a different geometric setting; the electrostatic theorem presupposes a static electro-vacuum system. The checks were targeted metadata/abstract and stated-theorem checks, not an exhaustive novelty search or a full independent reading of all external proofs. No private corpus text or terminology was sent to public search services.

10 Concrete future work

1. Choose one smooth fixed-N, finite-R filled candidate. Bound u, d, χ, all cutoff derivatives, and the metric chart constants with interval arithmetic. Preserve each actual Jacobian term.

2. Implement an H^2-compatible trace discretization and a conforming H^1 divergence discretization. Obtain a full-space inverse enclosure, not only a finite matrix inverse. Compare the block bound with a direct nonsymmetric inf-sup or validated resolvent bound.

3. Construct an equilibrated flux and a computable all-center source growth bound. Adaptive spatial partitions may reduce the measure budget; their covering and intercell terms must be proved, not sampled.

4. Verify a stronger-space nonlinear ball, with positivity and the source floor throughout that ball. Determine experimentally and then rigorously how the enclosure deteriorates with N and R.

5. Add certified exterior asymptotic tails, charge flux, invariant ADM mass, and full-cut area bounds. Preserve the sequential limits in the source rather than assuming one uniform numerical estimate.

6. Audit the neutral enclosing-area theorem independently, then the maximal-slice and electrostatic hypotheses in the equality argument. This is separate geometric work. A small residual cannot shortcut it.

These are proposed tasks, not achieved results or assertions that the source theorem is false. The current deliverable is a bounded analytic and verification contribution with an explicit noncertification boundary.

A Exact reading scopes and private locators

The three routing files `_audit/corpus-core-map.txt`, `_audit/corpus-foundations-map.txt`, and `_audit/corpus-applications-map.txt` were read in full, as was applicability row 510. They selected the passages below but supplied no proof evidence themselves. The user-provided confidentiality instruction, root study instructions and contribution policy were read. No additional AGENTS file was found in the checked task/source locations.

Pinned public source. Full scientific reads covered `build/sections/introduction.tex`, `statements.tex`, `cuts.tex`, `end_preparation.tex`, `filled_system.tex`, `a_priori.tex`, `elliptic_solver.tex`, `comparison.tex`, `variation.tex`, `stationary_region.tex`, and `rigidity.tex` (the latter ten names share the same sections directory). Assembly, `build/preamble.tex`, `build/references.bib`, and README were read in full. Chunked reads repaired output truncation; symbol-font support code was not treated as scientific text. No source PDF or unavailable figure was claimed read. External papers listed in Section 2.5 were read only to the public-check extent specified above, not in full. The neutral companion is used through the exact dependency statement in M510, not represented as independently read and proved here.

Private root conventions. In the following locators, D means `C:/Users/mural/OneDrive/Documents/`. Every locator denotes an actual local TeX passage, not a map summary. Input hashes preserve its version at authorship.

1. D`RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`, lines 53–395: elliptic `thm:coercive`, weak operator domain, boundary lifts, Neumann compatibility, Green realization, semilinear and complete quasilinear derivatives. Transfer: frozen block and weak source conventions. Limitation: its Hermitian coercive example is not the full M510 operator.

2. D`RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, full file: `thm:pert`, `eq:weak`, `eq:boundaryweak`, `thm:ball`, `eq:radii`, `prop:transfer`. Transfer: keep coefficient derivatives, real boundary conditions, bounded inverse and complete residual. Limitation: no inverse or nonlinear ball is supplied merely by selecting a finite coordinate representation.

3. *D*RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex, lines 145–184 and 450–569: finite moments and physical basis, thm:hypercircle and the source-error estimate. Transfer: physical norms, equilibrated flux, negative-source residual. Limitation: finite point jets need not be continuous on H^1; symmetry and flux hypotheses must be checked before a hypercircle argument.

4. *D*RamPatsCore50-TRs/TR-core-pat-8/TR-core-pat-8.tex, lines 221–520: weak realization, thm:coercive, eq:infsup, restricted Gram residual and complement, flux majorant, output correction, and thm:ball. Transfer: tested moments are lower diagnostics and onto-ness is not an automatic consequence of a one-sided lower bound. This mathematical reading is not a patent-filing task or a patentability opinion.

5. Study-root M517-Local-Penrose/TR-M517-Ram-Local-Penrose.tex, lines 1–160 deliberately reread for contextual conventions; additional displayed portions were inspected but are not used as theorem evidence. M517 treats fixed-seed Schwarzschild–anti-de Sitter perturbations and local jet residuals. Its setting is not the asymptotically Euclidean charged theorem here, and its C3 assessment is not inherited.

B Reproduction and delivery contract

The standalone TeX, validation/provenance scripts and PDF are confined to M510-Charged-Penrose/; author audit files have prefix _audit/M510 except the designated contribution assessment JSON. The build uses only _build/M510-author/. Source corpus, catalogue, queue, progress, shared delivery and other reports are not edited. The full scientific TeX is released for parent review before testing and physical export. Final receipts, assessment and frozen manifest refer to the final source hash. No parent-thread messaging is used.

Run exact checks with py-3.10 and the accompanying validation script. The existing export helper is invoked with the absolute source path, --round1 (or the permitted second minor-format round), and --build-tagM510-author, using the prescribed bundled Python and installed MiKTeX. Native editor opening and compilation are attempted first; unavailable tools or platform-directory failures must be recorded as observed, never as successful native compilation. All final PDF pages are rendered and visually inspected. The compact status and frozen manifest identify the resulting build, proof checks, test scope, page inspection and immutable input/output hashes. These operational checks do not expand the mathematical claims of this report.

References

[1] OpenAI. *The spacetime Penrose inequality with charge and original-data rigidity*. October 5, 2026. Pinned source and module locators in Sections 1–2.5.

[2] OpenAI. *The spacetime Penrose inequality and enclosing area*. September 27, 2026. Definitions 1.1–1.2 and Theorem 1.3, as explicitly invoked by M510; not independently certified here.

[3] M. Khuri, G. Weinstein, S. Yamada. Proof of the Riemannian Penrose inequality with charge for multiple black holes. *J. Differential Geom.* 106 (2017), 451–498. https://arxiv.org/abs/1409.3271.

[4] W. Littman, G. Stampacchia, H. F. Weinberger. Regular points for elliptic equations with discontinuous coefficients. *Ann. Scuola Norm. Sup. Pisa* (3) 17 (1963), 43–77. https://numdam.org/item/ASNSP_1963_3_17_1-2_43_0/.

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[7] D. Gallistl. Variational formulation and numerical analysis of linear elliptic equations in nondivergence form with Cordes coefficients. *SIAM J. Numer. Anal.* (2017). https://arxiv.org/abs/1606.05631; https://doi.org/10.1137/16M1080495.

[8] M. Subbarao. Private Ram technical reports: TR8, TR1 stability, revised TR250, and core TR8. Exact local files, selected passages, transfer conditions, and limitations in Appendix A. No public availability or external publication status is asserted.